# Astha Mishra

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## Education

**Bachelor of Technology,** *Sister Nivedita University* 2021 – 2025 | Kolkata, India
Computer Science & Engineering (CGPA: 8.3)
Key Courses: Machine Learning, Data Structures, Operating Systems

## Skills

Python | React.js | Scikit-learn | NumPy | Pandas | MySQL/SQL | Git | Docker

## Professional Experience

**Machine Learning Intern,** *Indian Space Research Organisation (ISRO)* 09/2024 – 05/2025 | Kolkata, India

- Designed and implemented a scalable Python-based data processing pipeline for multi-temporal SAR datasets, achieving an $R^2$ of 0.87 for Above-Ground Biomass (AGB) estimation in the Indian Sundarbans
- Built a modular preprocessing framework with noise reduction and normalization components, improving data consistency by 30%
- Engineered reusable feature extraction modules to compute 18+ polarimetric descriptors and applied dimensionality reduction techniques (PCA, RFE) for optimized performance
- Developed and optimized machine learning models (Random Forest, XGBoost, SVR) for predictive analytics on high-dimensional geospatial datasets
- Created synthetic data generation workflows to improve model robustness and generalization in data-scarce scenarios
- Designed visualization pipelines using Matplotlib and Seaborn for performance monitoring and validation
- Performed exploratory data analysis to identify trends and anomalies, improving model feature selection.
- Collaborated with cross-functional teams to translate analytical requirements into structured datasets and insights.

## Projects

**Yoga Posture Detection System,** *AI + Web App*

- Developed a real-time yoga posture detection web application for accurate pose classification with a split-screen interface displaying live webcam feedback alongside dynamic, timed pose instructions.
- Used TensorFlow and OpenCV in Python, integrating deep learning models like PoseNet, MobileNet, ResNet, and OpenPose.
- Used GitHub Issues for bug tracking and feature management

**Chatify,** *React.JS*

- Developed a real-time chat application for bidirectional communication for seamless user interaction and real-time synchronization across clients.
- Implemented socket programming with a responsive UI and dynamic message updates.

**StockStream Pipeline – Dockerized Backend Data System**

- Built a containerized backend data pipeline using Python and Docker for real-time stock data processing
- Designed modular components following microservices principles for scalable data ingestion and transformation
- Implemented automated workflows showcasing distributed systems and backend architecture concepts
- Implemented caching using Redis to optimize data retrieval

## Publications

**Estimating Biophysical Parameters of Mangrove Forests of Indian Sundarbans using Extreme Gradient Boosting with C-band SAR data and Ground Measurements,** *Tanumi Kumar, Astha Mishra, Sharmistha B Pandey, Prabir Kumar Das, A. O. Varghese* (2025). *Abstract accepted in ISG-ISRS* 2025 *National Symposium* (*Abstract ID*: 55).